

chapter-1

Introduction to op amplifier

An operational amplifier is a high gain electronic voltage amplifier that has two input and one output. It amplifies the difference between the input voltages and produces a single output voltage.

$$\Rightarrow V_{out} = A (V_+ - V_-)$$

$V_+ \Rightarrow$ non inverting input

$V_- \Rightarrow$ inverting input

$A \Rightarrow$ open loop gain (its value may be probably $10^5 - 10^6$)

If $V_+ > V_- \Rightarrow$ output positive then V_{out}

$V_+ < V_- \Rightarrow$ output negative then V_{out}

Example:

$$v_+ = 2.0001 \text{ V}$$

$$v_- = 2.000 \text{ V}$$

$$d^2 (v_+ - v_-)$$

$$= (2.0001 - 2.000) \text{ V}$$

$$= 0.0001 \text{ V}$$

if the gain is 10^5 then

$$v_{out} = A_d$$

$$= 10^5 \times 0.0001 \text{ V}$$

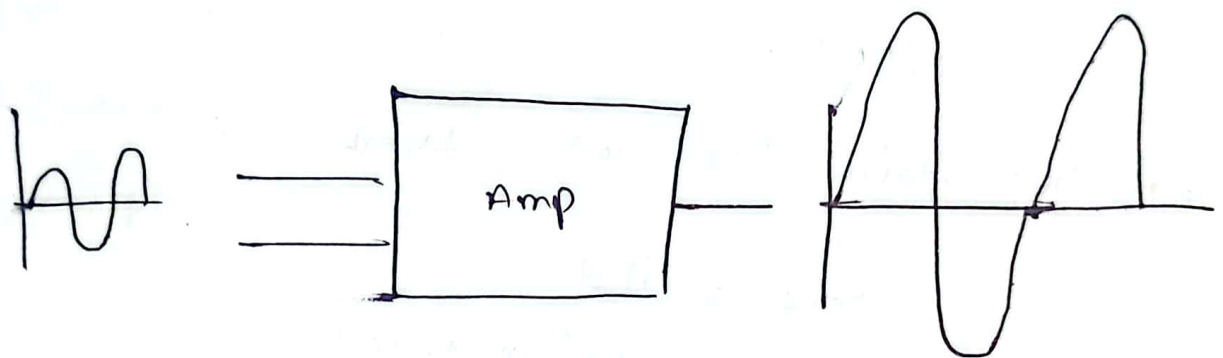
$$= 1 \text{ V}$$

operational Amplifier can be used

- ① to detect the small signal
- ② to reduce noise
- ③ to measure the value accurately
- ④ it's easy to process signal.

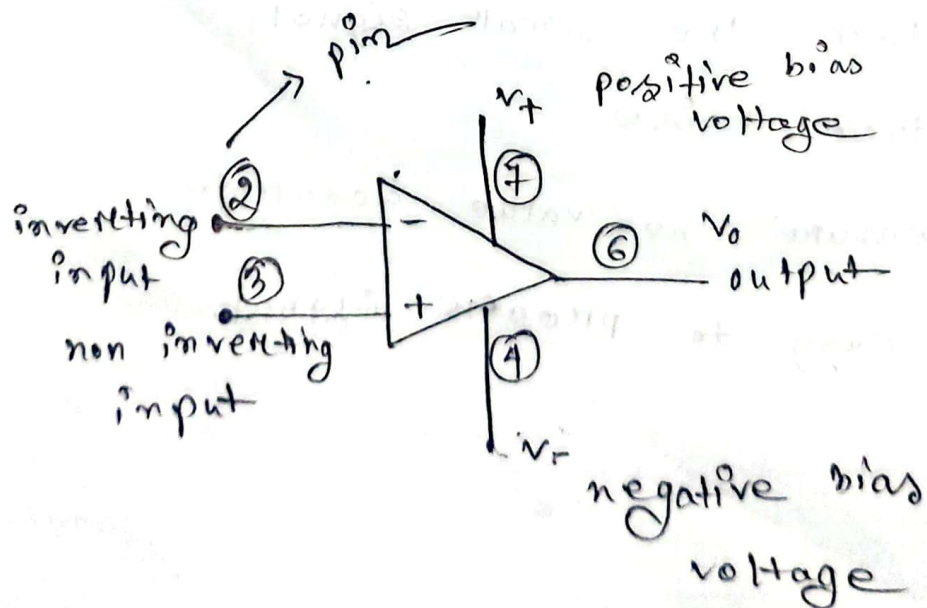
mathematical operation:

- ① Addition
- ② subtraction
- ③ integration
- ④ differentiation

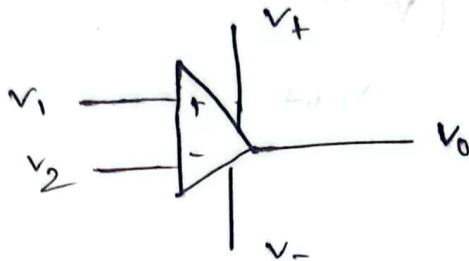


circuit symbol

pin no of Amp is 741, which is 8 terminal. pin no 2 is terminal.

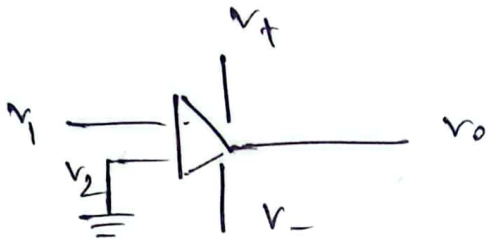


$\Rightarrow 1, 5$ pin $\Rightarrow$ we offset
 $\Rightarrow 8$ pin $\Rightarrow$ not connected

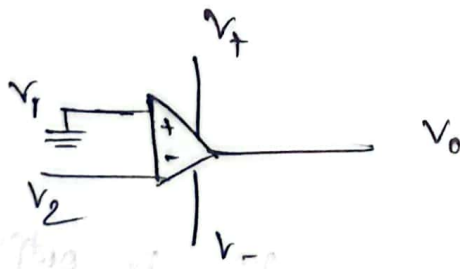
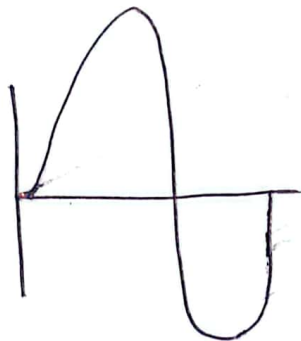


Differential Amplifier

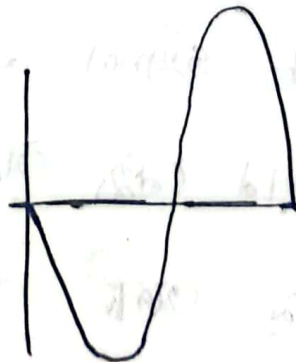
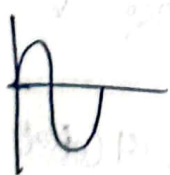
$$V_{out} = A (V_1 - V_2)$$



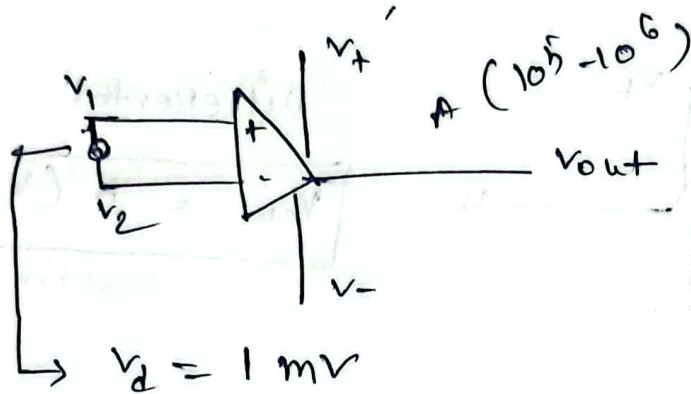
$$\begin{aligned} V_{out} &= A (V_1 - V_2) \\ &= A (V_1 - 0) \quad [V_2 = 0 \text{ or } V_2 = Gnd] \\ &= A V_1 \end{aligned}$$



$$\begin{aligned} V_{out} &= A (V_1 - V_2) \\ &= A (0 - V_2) \\ &= -A V_2 \quad (- \text{ sign means } 180^\circ \text{ phase shift}) \end{aligned}$$



ଉଦାହରଣ v_+ ଓ v_- ସମ୍ପର୍କିତ କ୍ଷମ୍ପା ଉଦାହରଣ (ଉଦା ୨)



$$A = 10^5$$

$$v_{out} = 1 \text{ mV} \times 10^5$$

$$= 100 \text{ V}$$

Another example

$$v_d = 1 \text{ V}$$

$$A = 10^5$$

$$v_{out} = 1 \times 10^5$$

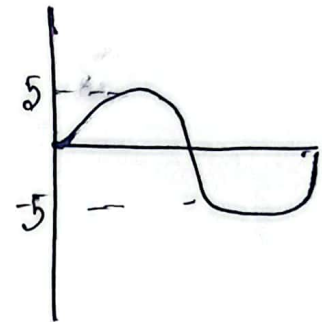
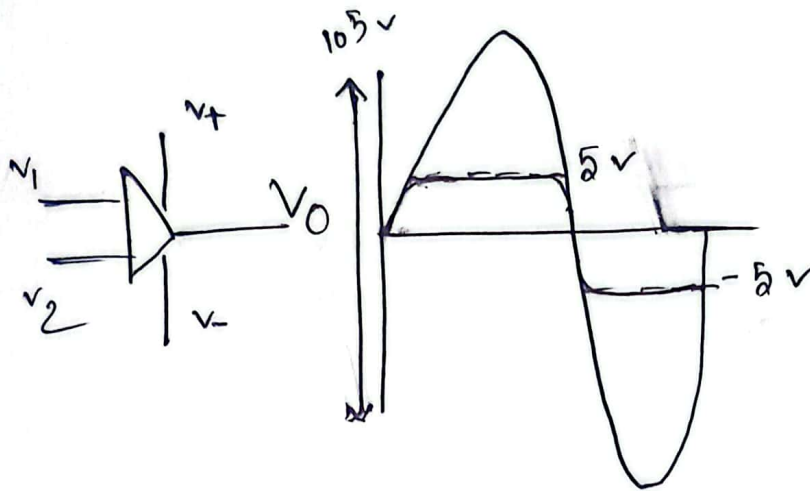
$$= 100000 \text{ V}$$

$$= 10^5 \text{ V}$$

ଯେ ଚାହୁଁଛନ୍ତି output signal ଯାହା ଏହା output signal ଓ ଏହା ଫଳାଫଳ ନିଶ୍ଚିତ ଭାବେ v_+ ଓ v_- ସମ୍ପର୍କିତ କ୍ଷମ୍ପା ଉଦାହରଣ (ଉଦା ୨) ଉପରେ ନିର୍ଭର କରେ।

21/02/2021

Suppose $v_+ = 5V$
 $v_- = -5V$



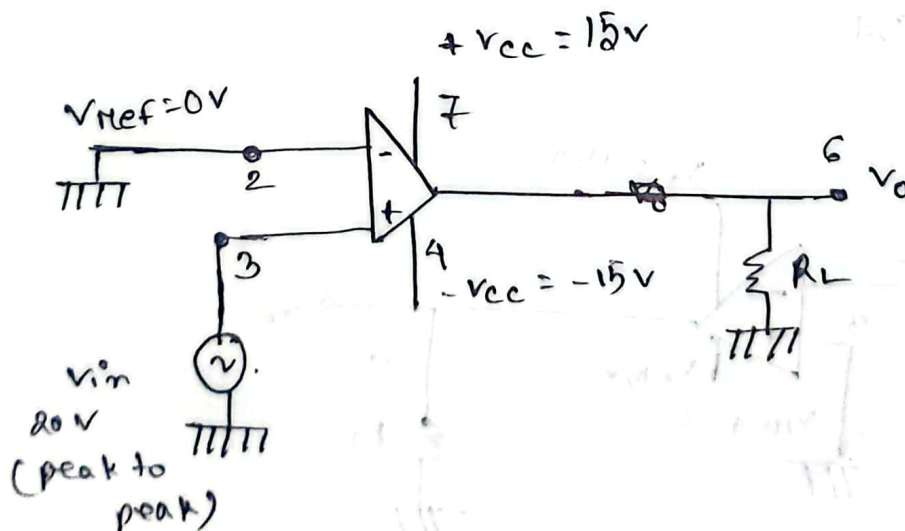
2.2.1 open loop voltage gain:

open loop voltage gain is the ratio of the output voltage to the differential input voltage of an operational amplifier when no feedback is connected.

$$A_{OL} = \frac{V_{out}}{V_+ - V_-} = \frac{V_o}{E_d}$$

2.3.1 zero crossing detector

1. Non inverting:



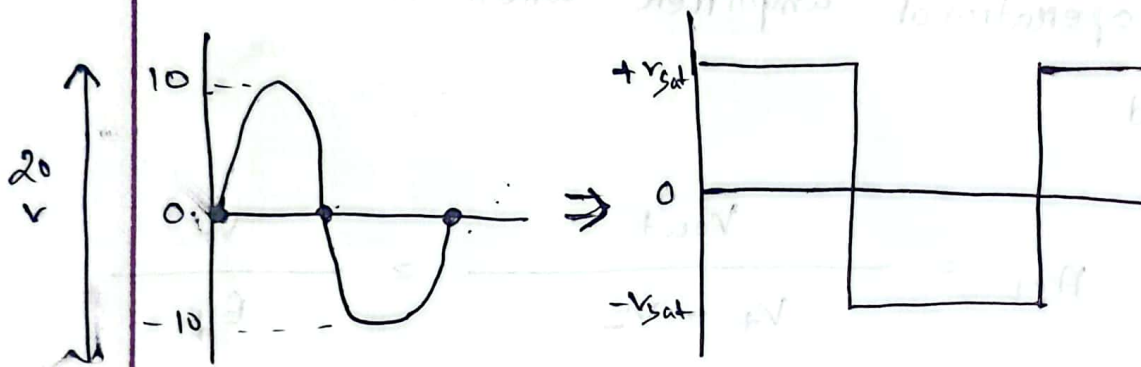
non inverting: when V_{in} is above V_{ref}

$$V_o = +V_{sat}$$

$$v_o = A(v_+ - v_-)$$

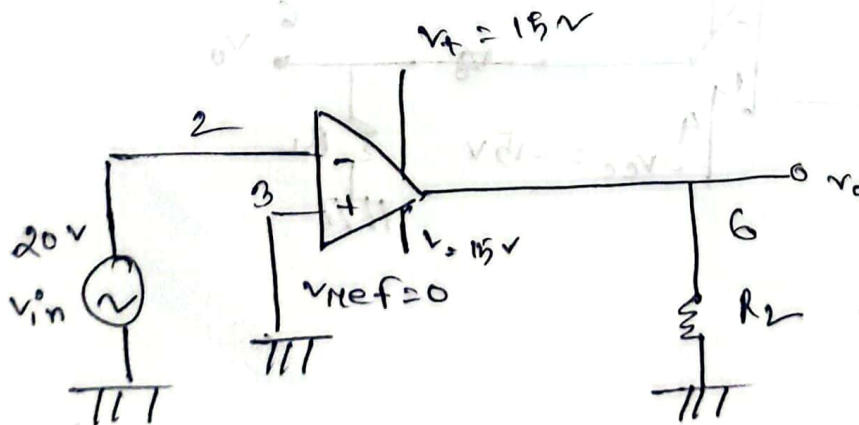
$$\geq A(v - 0)$$

$$\geq +Av$$



\$\Rightarrow\$ zero crossing detector means when it detects 0, it changes his position.

2. inverting:

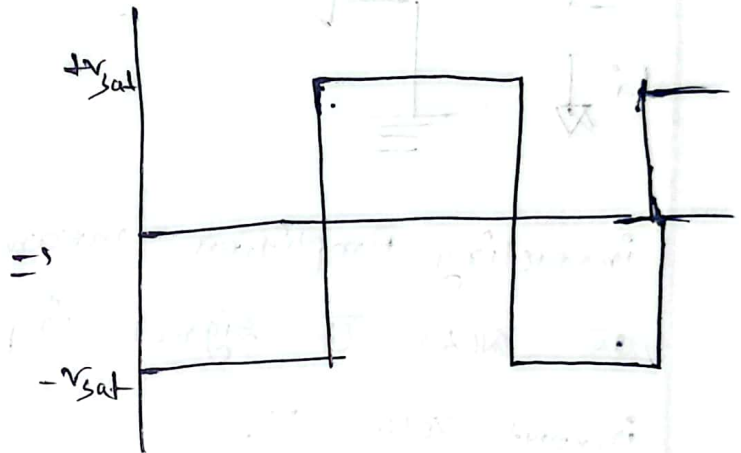
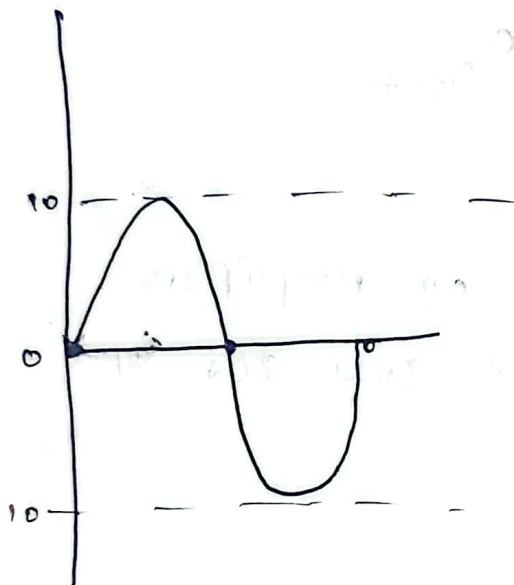


Inverting: when \$v_{in}\$ is above \$v_{ref}\$
 $v_o = -V_{set}$

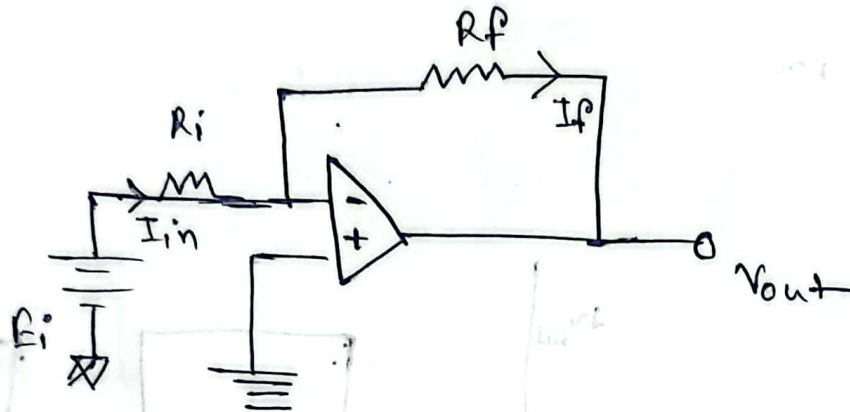
$$v_o = A(v_+ - v_-)$$

$$= A(0 - x)$$

$$= -A x$$



3.1 Inverting Amplifier



Inverting Amplifier - means op Amplifier

ଓପ୍ ଆମ୍ପ୍ଲିଫିର ଯାହା ଇନ୍ପୁଟ ଲେଭେଲ୍ ଓ
ଇନ୍ଭେରଟ୍ କରେ, ତାହା ଇନ୍ଭେରଟିଂ ଆମ୍ପ୍ଲିଫିର।



Virtual ground concept:

ଓପ୍ ଆମ୍ପ୍ଲିଫିର ଇନ୍ପୁଟ ଓ ଆଉଟପୁଟ
ବିପରୀତ ଚିହ୍ନରେ ହେବ, ଯଦି ଇନ୍ପୁଟ ଧନାତ୍ମକ ହେବ, ତେବେ ଆଉଟପୁଟ ଋଣାତ୍ମକ ହେବ।

$$\text{Voltage gain } A_v = \frac{V_{out}}{V_{in}}$$

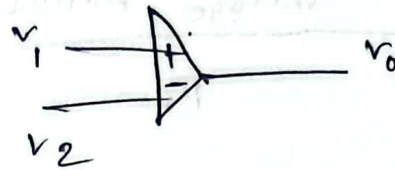
$$\Rightarrow V_{out} = A_v \times V_{in}$$

$$\Rightarrow 5 = 10^5 \times V_{in}$$

$$\Rightarrow V_{in} = \frac{5}{10^5}$$

$$= 0.00005$$

$$\therefore V_d \approx 0$$

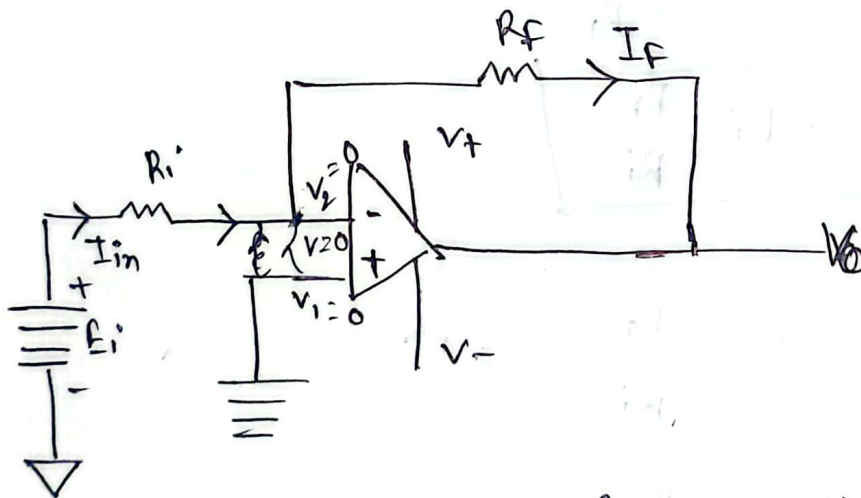


$$V_{in} = V_1 - V_2 = V_d$$

$$V^+ - V^- = 0$$

$$\Rightarrow V^+ = V^-$$

अतः $V^+ = 0$ अतः $V^- = 0$



ohm's law, $I_{in} = \frac{\text{voltage across in } R_i}{R_i}$

$$= \frac{E_i - 0}{R_i} = \frac{E_i}{R_i}$$

$$I_f = \frac{\text{voltage across } R_f}{R_f}$$

$$= \frac{0 - V_o}{R_f}$$

$$= - \frac{V_o}{R_f}$$

According to KCL

$$I_{in} = I_f$$

$$\Rightarrow \frac{E_i}{R_i} = - \frac{V_o}{R_f}$$

$$\Rightarrow \boxed{V_o = - E_i \frac{R_f}{R_i}}$$

$$\Rightarrow \frac{V_o}{E_i} = - \frac{R_f}{R_i}$$

$$\Rightarrow A_v = - \frac{R_f}{R_i}$$

$$\Rightarrow \boxed{\text{voltage gain } A_v = \frac{V_o}{E_i} = - \frac{R_f}{R_i}}$$

Example: 3.1

$$R_F = 100 \text{ k}\Omega$$

$$R_i = 10 \text{ k}\Omega$$

$$E_i = 1 \text{ V}$$

$$\begin{aligned} \textcircled{a} \quad I &= \frac{E_i}{R_i} = \frac{1}{10} \\ &= 0.1 \text{ mA} \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad v_o &= -E_o \times \frac{R_F}{R_i} \\ &= -1 \times \frac{100}{10} \\ &= -10 \text{ V} \end{aligned}$$

$$\begin{aligned} \textcircled{c} \quad A_{cl} \text{ (voltage gain)} &= -\frac{R_F}{R_i} \\ &= -\frac{100}{10} \\ &= -10 \end{aligned}$$

Another equation

$$\begin{aligned} A_{cl} &= \frac{v_o}{E_i} = \frac{-10}{1} \\ &= -10 \text{ V} \end{aligned}$$

3.1.3 load and output current

the load current I_L that flows through R_L is determined by R_L and v_o .

$$\text{Load current } I_L = \frac{v_o}{R_L}$$

$$\text{output current } I_o = I + I_L$$

Example 3.2

$$I = 0.1 \text{ mA}$$

$$v_o = -10 \text{ V}$$

$$R_L = 25 \text{ k}\Omega$$

$$\begin{aligned} \textcircled{a} \quad I_L &= \frac{v_o}{R_L} \\ &= -\frac{10}{25} \\ &= -0.4 \text{ mA} \end{aligned}$$

$$\begin{aligned} \textcircled{b} \quad I_o &= I + I_L \\ &= 0.1 + 0.4 \\ &= 0.5 \text{ mA} \end{aligned}$$

Ex 3.3

$$R_f = 250 \text{ k}\Omega$$

$$R_i = 10 \text{ k}\Omega$$

$$E_i = -0.5 \text{ V}$$

$$\textcircled{a} \quad I = \frac{E_i}{R_i} = \frac{0.5}{10 \text{ k}\Omega} = 0.005 \text{ mA}$$

$\textcircled{b}$ the voltage across R_f

$$V_f = I R_f$$

$$= 0.005 \times (250 \text{ k}\Omega)$$

$$= 12.5 \text{ V}$$

$$\textcircled{c} \quad V_o = - E_i \times \frac{R_f}{R_i}$$

$$= - (-0.5) \times \frac{250}{10}$$

$$= 12.5 \text{ V}$$

Ex 3.4

$$I_L = 2 \text{ mA}$$

$$\textcircled{a} \quad I_L = \frac{V_o}{R_L}$$

$$\Rightarrow R_L = \frac{V_o}{I_L} \Rightarrow \frac{12.5}{2} = 6.25 \text{ k}\Omega$$

$$\textcircled{b} I_0 = I + I_L$$

$$= 0.05 + 2 \text{ mA}$$

$$= 2.05 \text{ mA}$$

③ the circuit input resistance $R_i = 10 \text{ k}\Omega$

Ex 3.5

$$R_f = 20 \text{ k}\Omega$$

$$R_i = 10 \text{ k}\Omega$$

$$\textcircled{a} \text{ voltage gain } A_{cl} = - \frac{R_f}{R_i}$$

$$= - \frac{20 \text{ k}\Omega}{10 \text{ k}\Omega}$$

$$= -2 \text{ k}\Omega$$

Ex 3.6

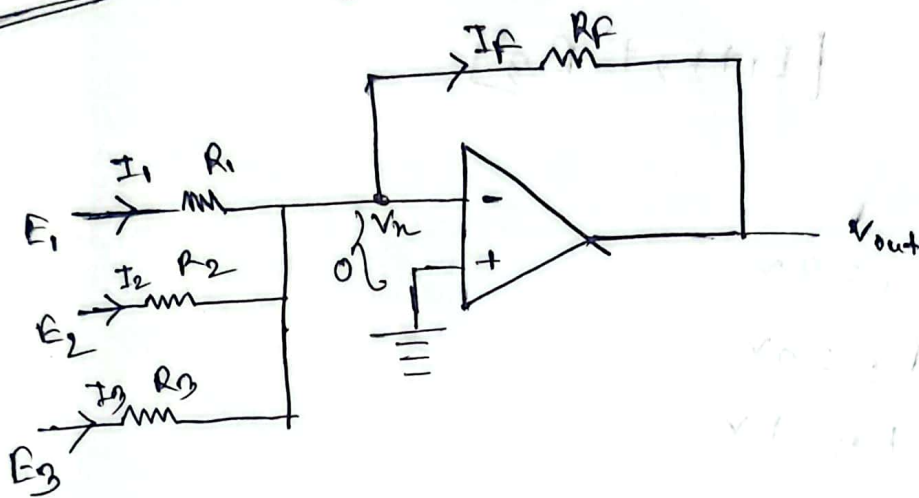
Input voltage $E_i = -5 \text{ V}$

$$V_o = - \frac{R_f}{R_i} \times E_i$$

$$= - \frac{20}{10} \times (-5 \text{ V})$$

$$= 10 \text{ V}$$

3.2.1. Inverting Adder:



using KCL into V_n ,

input current = output current

$$\Rightarrow I_1 + I_2 + I_3 = I_F$$

$$\Rightarrow \frac{E_1 - V_n}{R_1} + \frac{E_2 - V_n}{R_2} + \frac{E_3 - V_n}{R_3} = \frac{V_n - V_{out}}{R_F}$$

$$\Rightarrow \frac{E_1 - 0}{R_1} + \frac{E_2 - 0}{R_2} + \frac{E_3 - 0}{R_3} = \frac{-V_{out}}{R_F}$$

$$\Rightarrow \frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3} = -\frac{V_{out}}{R_F}$$

$$\Rightarrow V_{out} = - \left[\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3} \right] \times R_F$$

If $R_1 = R_2 = R_3 = R$ then

$$V_{out} = -\frac{R_F}{R} [E_1 + E_2 + E_3]$$

if $R_F = R$ and $\overline{v_{out}}$

$$v_{out} = - [E_1 + E_2 + E_3]$$

Ex: 3.8

$$E_1 = 2V$$

$$E_2 = 3V$$

$$E_3 = 1V$$

$$R = 10k\Omega$$

$$v_{out} = - [E_1 + E_2 + E_3]$$

$$= - [2 + 3 + 1]$$

$$= -6V$$

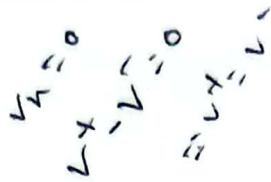
Ex: 3.9

E_3 is reversed $E_3 = -1V$

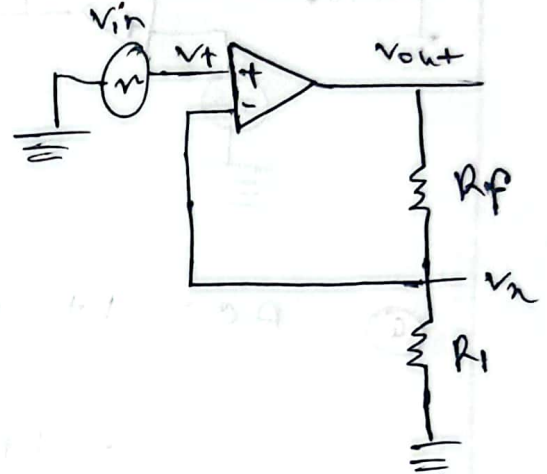
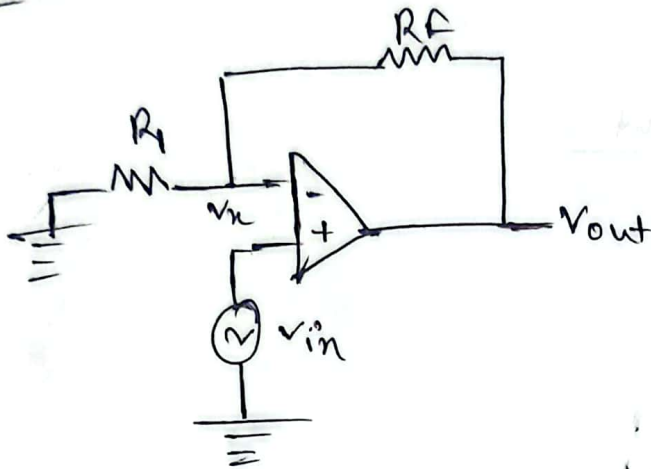
$$v_{out} = - [E_1 + E_2 + E_3]$$

$$= - [2 + 3 - 1]$$

$$= -4V$$



3.5 non inverting Amplifier



$$V_n = \frac{R_1}{R_1 + R_f} \times V_{out}$$

$$\Rightarrow V^+ = V^- = V_n = \frac{R_1}{R_1 + R_f} \times V_{out}$$

$$\Rightarrow V_{in} = \frac{R_1}{R_1 + R_f} \times V_{out}$$

$$\Rightarrow V_{out} = \left(\frac{R_1 + R_f}{R_1} \right) \times V_{in}$$

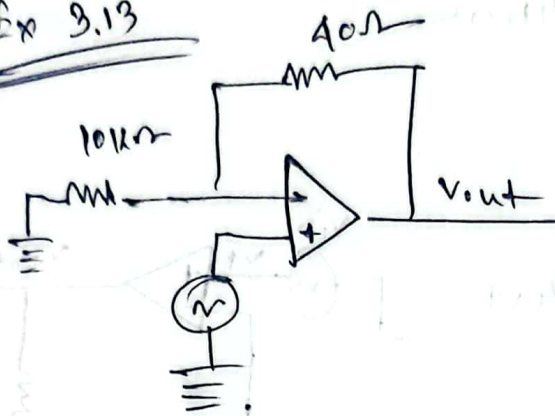
$$\Rightarrow \frac{V_{out}}{V_{in}} = \left(1 + \frac{R_f}{R_1} \right)$$

$$\Rightarrow A = \left(1 + \frac{R_f}{R_1} \right)$$

$$V_{Rf} = I R_f$$

$$= \frac{R_1}{R_1} \times R_f$$

Ex 9.13



(a) $AC = 1 + \frac{R_F}{R_f}$

$$= 1 + \frac{40}{10}$$

$$= \frac{10 + 40}{10}$$

$$= 5$$

Ans: 3.15

(a)

$$V_o = V_i = 4V$$

(b)

$$I_L = \frac{V_o}{R_L} = \frac{4}{10}$$

$$= 0.4mA$$

(c)

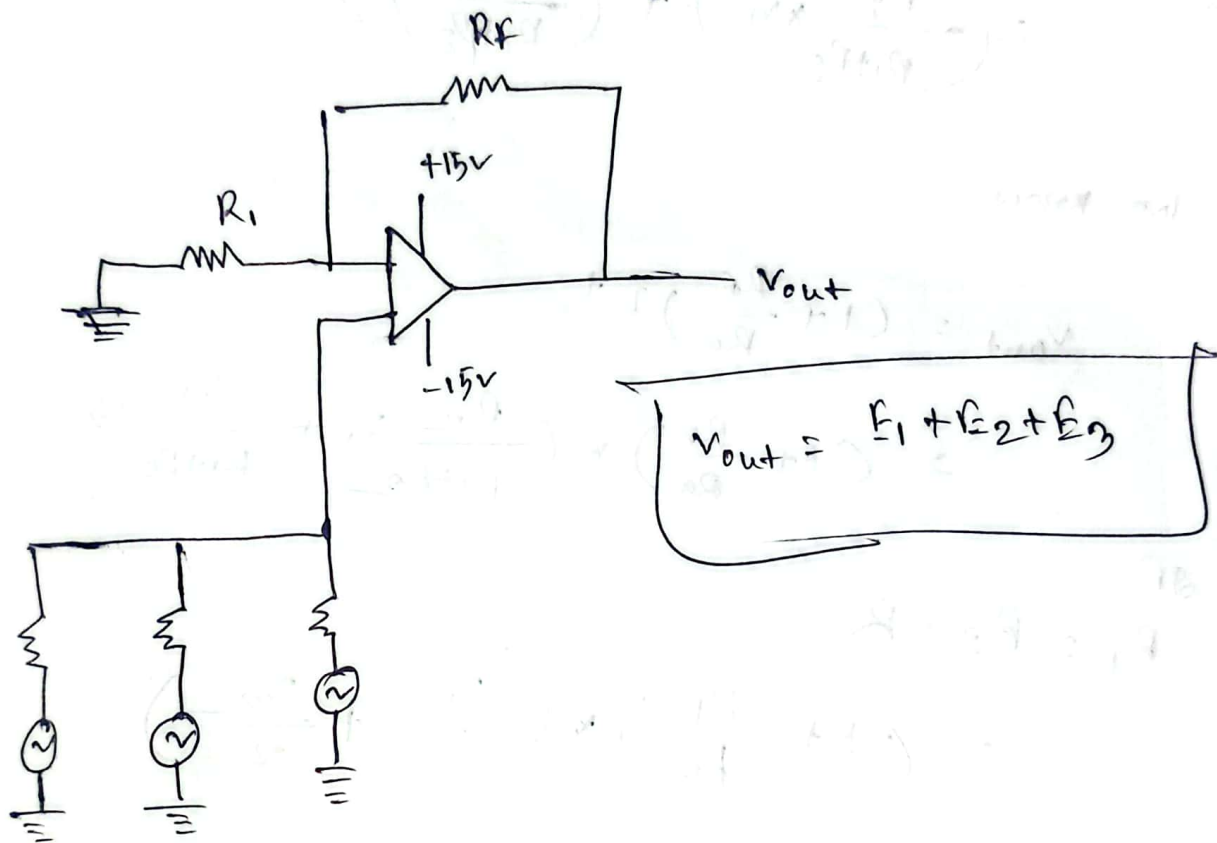
$$I_o = I + I_L$$

$$I_o = 0 + I_L$$

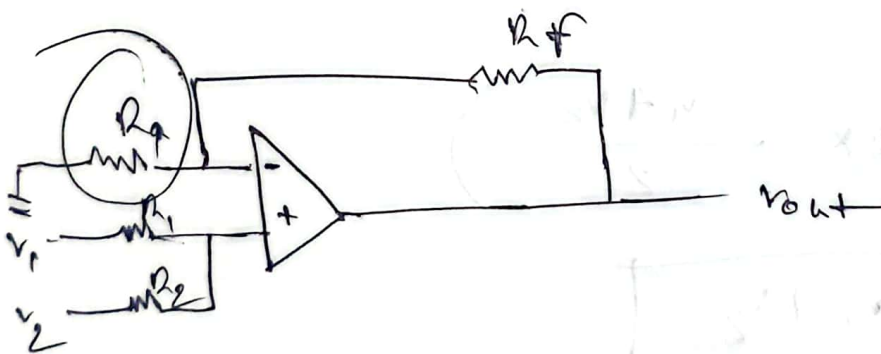
$$= 0 + 0.4 = 0.4mA$$

(Ideal input $I_{in} = 0$)

non inverting adder:



$$V_{out} = E_1 + E_2 + E_3$$



$$V_{out} = \left(1 + \frac{R_F}{R_A}\right) V^+$$

using voltage divider rule

$$V_1^+ = \frac{R_2}{R_1 + R_2} \times V_1$$

$$V_2^+ = \frac{R_1}{R_1 + R_2} \times V_2$$

$$v^+ = v_1^+ + v_2^+$$

$$= \left(\frac{R_2}{R_1 + R_2} \right) v_1 + \left(\frac{R_1}{R_1 + R_2} \right) v_2$$

We know

$$v_{out} = \left(1 + \frac{R_f}{R_a} \right) v^+$$

$$= \left(1 + \frac{R_f}{R_a} \right) \times \left(\frac{R_2}{R_1 + R_2} v_1 + \frac{R_1}{R_1 + R_2} v_2 \right)$$

if

$$R_1 = R_2 = R$$

$$= \left(1 + \frac{R_f}{R_a} \right) \times \left(\frac{v_1}{2} + \frac{v_2}{2} \right)$$

if

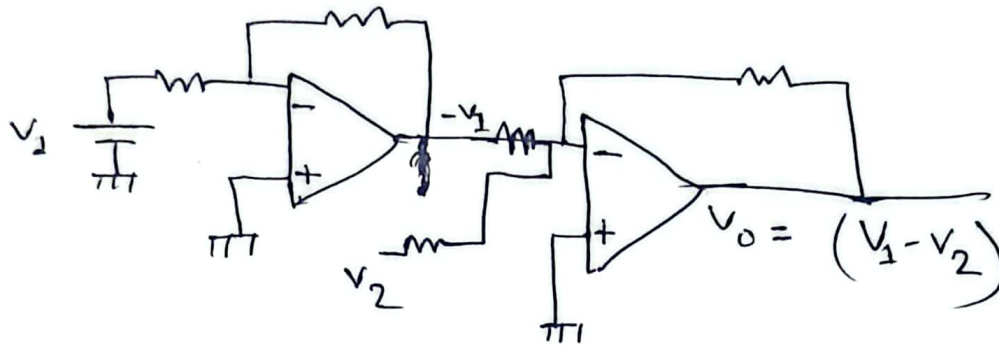
$$R_f = R_a$$

$$= \left(2 \times \frac{v_1 + v_2}{2} \right)$$

$$v_{out} = v_1 + v_2$$

Subtractor:

Inverting $\Rightarrow$



Sub $\Rightarrow (V_1 - V_2)$

